

Gases 5

Essential Pre-Uni Chemistry B3.5

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** GCSE C2, A Level P1

RTP = room temperature and pressure.

Any gas occupies 24 dm^3 per mole at RTP.

Avogadro's number, $N_A = 6.02 \times 10^{23}$.

Part A

(a)

Calculate the mass of 1.0 m^3 of neon at RTP.

Part B

(b)

Calculate the mass of 20 cm^3 of $(\text{CH}_3)_2\text{O}$ at RTP.

Part C

(c)

Calculate the mass of 420 cm^3 of ammonia at RTP. Give your answer to 2 significant figures.

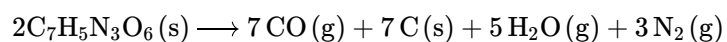
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TNT

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level P2

TNT is used as an explosive. It can decompose according to the following equation:



Part A

RMM

Calculate the relative molecular mass of TNT, rounding your answer to an integer.

Part B

Moles of gas

The volume of gas produced at 400°C , when 10 g of TNT explode, is to be calculated.

How many moles of gas are produced from 1 mol of TNT?

Part C

Volume of gas

At 400°C and 1 atm, 1 mol of gas occupies 55 dm^3 .

Calculate the volume of gas produced under these conditions from 10 g of TNT.

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Compounds TBC

Subject & topics: Chemistry | Foundations | Stoichiometry **Stage & difficulty:** A Level C2

When calcium oxide is heated with carbon, an ionic compound, **D**, containing 62.5% of calcium and 37.5% of carbon (by mass), is formed. Under similar conditions, aluminium metal and carbon produce compound **E** which contains 75% of aluminium and 25% of carbon.

When treated with cold water:

compound **D** produces a gaseous hydrocarbon **F** containing 92.3% of carbon

compound **E** produces another gaseous hydrocarbon **G** containing 75% of carbon

Part A

D

Determine the empirical formula of compound **D**.

Part B

E

Determine the empirical formula of compound **E**.

Part C

F

Determine the empirical formula of compound **F**.

Part D

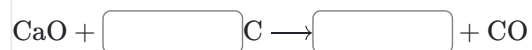
G

Determine the empirical formula of compound **G**.

Part E

Reaction to form D

Write a balanced equation for the reaction of calcium oxide with carbon, using the empirical formula for **D** you have previously deduced.



Items:

Part F

Reaction to form E

Write a balanced equation for the reaction of aluminium metal and carbon to form **E** (do not include state symbols).

Part G

Reaction of E with water

Assuming the empirical formula you deduced for **G** is also its molecular formula, write a balanced equation for the reaction when compound **E** is treated with water.



Items:

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Lattice Enthalpy Definition

Subject & topics: Chemistry | Inorganic | Bonding & IMFs**Stage & difficulty:** A Level P1**Part A****Lattice enthalpy definition**

Fill in the missing words:

Lattice enthalpy of formation $\Delta_{LE}H$ is the energy change when one of an ionic solid is formed from its .

Lattice formation enthalpies are always . The magnitude of the lattice enthalpy is affected by both the ionic and the on the ions.

Items:

Part B

Lattice energy

For which compound is the lattice energy likely to have the greatest numerical value (i.e. the greatest magnitude, disregarding the sign)?

- ☐ lithium fluoride
- ☐ lithium iodide
- ☐ rubidium chloride
- ☐ sodium chloride

Part A adapted with permission from OCSEB, A-Level Chemistry, June 1995, Special Paper, Question 4;

Part B adapted with permission from UCLES, A-Level Chemistry, November 1995, Paper 4, Question 7

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Lattice Enthalpy Estimation

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level P1

Within the ionic model, lattice enthalpies in kJ mol^{-1} may be estimated using the equation:

$$\Delta_{\text{L}} H^{\ominus} = \frac{C \cdot z^{+} \cdot z^{-} \cdot \nu}{(r^{+} + r^{-})} - 2.5\nu$$

Where:

C is a constant approximately equal to 105 000 units;

z^{+} and z^{-} are the *signed* charges on the cation and anion respectively in units of e ;

ν is the number of ions in the formula (e.g. 3 for MgI_2);

r^{+} and r^{-} are the radii of the ions in pm;

The -2.5ν term corrects for the difference between internal energy and enthalpy.

The table below shows the radii for certain ions.

Ion	Li^{+}	Na^{+}	Ca^{2+}	Cr^{3+}	Hg^{+}	O^{2-}	F^{-}	Cl^{-}	Br^{-}
Radius / pm	74	102	100	62	158	140	133	180	195

Estimate the values of $\Delta_{\text{L}} H^{\ominus}$ for the following compounds, using the equation given. Give your answers to 3 significant figures.

Part A

LiBr

$\Delta_{\text{L}} H^{\ominus}$ for LiBr

Part B
 Na_2O

$\Delta_{\text{L}}H^\circ$ for Na_2O

Part C
 CaF_2

$\Delta_{\text{L}}H^\circ$ for CaF_2

Part D
 Cr_2O_3

$\Delta_{\text{L}}H^\circ$ for Cr_2O_3

Part E
 Hg_2Cl_2

$\Delta_{\text{L}}H^\circ$ for Hg_2Cl_2

Part F

Poor approximation

Experimentally found lattice enthalpies are:

Lattice	LiBr	Na ₂ O	CaF ₂	Cr ₂ O ₃	Hg ₂ Cl ₂
$\Delta_L H^\ominus / \text{kJ mol}^{-1}$	−800	−2530	−2635	−15115	−1950

For which compound is the ionic model a poor approximation?

Based on question F4.4 from the Physical Chemistry book

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Bonding in Ozone

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C2

A molecule of ozone is an allotrope of oxygen containing three oxygen atoms covalently bonded together, rather than the usual diatomic form we are more familiar with.

Part A

Valence electrons

How many valence electrons does a molecule of ozone contain?

Part B

Highly symmetric

One theoretically predicted, but never observed structure of ozone would mean all atoms in the form would be equivalent. What would this structure look like?

Part C

Actual geometry

In practice, ozone is a bent (V-shaped) molecule with a bond angle of about 117° . Draw a dot-and-cross diagram that accounts for the number of valence electrons present and satisfies the octet rule. Use your diagram to fill in the blanks below.

The central oxygen atom in the structure is forming bonds and has lone pair(s). It holds a formal charge. One of the end atoms holds a formal charge, and is bonded to the central atom by means of a bond. This end atom has lone pair(s) according to the structure drawn. The other end atom is not charged, forms a bond with the central atom, and has lone pair(s) according to the structure drawn.

Items:

Part D

Delocalised bonding

In reality, the two end oxygen atoms in ozone are equivalent and the π bonding is delocalised. If there are two π bonding electrons spread out across the three atoms, what is the overall bond order in each O—O link?

Shapes of Halide Compounds

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C1

For each of the following, deduce the shape of the molecules and enter a one to two word answer, using appropriate shape of molecule terminology, e.g. "linear".

Part A



Deduce the shape of BBr_3 .

Part B



Deduce the shape of PF_3 .

Part C



Deduce the shape of SF_4 .

Part D



Deduce the shape of IF_5 .

Part E



Predict the shape of the species formed from the reaction of AlCl_3 with Cl^- .

Adapted with permission from OCSEB, STEP Chemistry, Jun 1998, Question 4

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Shapes of Xenon Compounds

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C2

For each of the following, deduce the shape of the molecules and enter a one to two word answer, using appropriate shape of molecule terminology, e.g. "linear".

Part A



Describe the shape of XeF_2 .

Part B



Describe the shape of XeOF_2 .

Part C



Describe the shape of XeO_4 .

Part D



Describe the shape of XeF_4 .

Part E



Describe the shape of XeOF_4 .

Part A adapted with permission from OCR, STEP Chemistry, June 1999, Question 5

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Shapes of Molecules and Ions

Subject & topics: Chemistry | Inorganic | Bonding & IMFs**Stage & difficulty:** A Level P1**Part A**

By considering the number of lone and bonding pairs of electrons, predict the shape of F_2O .

Part B

By considering the number of lone and bonding pairs of electrons, predict the shape of H_3O^+ .

Part C

By considering the number of lone and bonding pairs of electrons, predict the shape of ClF_4^- .

Part D



Antimony, Sb, is in group 15 of the Periodic Table. It forms a series of salts which contain the SbF_5^{n-} anion, the structure of which is a square-based pyramid:

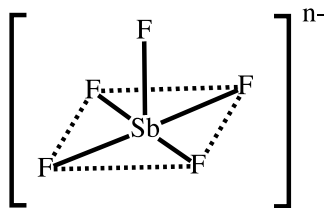


Figure 1: Structure of the SbF_5^{n-} anion

Deduce the total number of electrons around the antimony atom.

Deduce the value of n .

Adapted with permission from UCLES, A-Level Chemistry, June 1991, Paper 3, Question 2

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Shapes of Molecules and Ions Extension

Subject & topics: Chemistry | Inorganic | Bonding & IMFs **Stage & difficulty:** A Level C1

Antimony, Sb, is in group 15 of the Periodic Table. It forms a series of salts which contain the SbF_5^{n-} anion, the structure of which is a square-based pyramid:

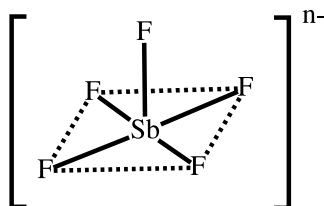


Figure 1: Structure of the SbF_5^{n-} anion

Deduce the oxidation number of Sb in the SbF_5^{n-} anion above.

Adapted with permission from UCLES A-Level Chemistry June 1991, Paper 3, Q2

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